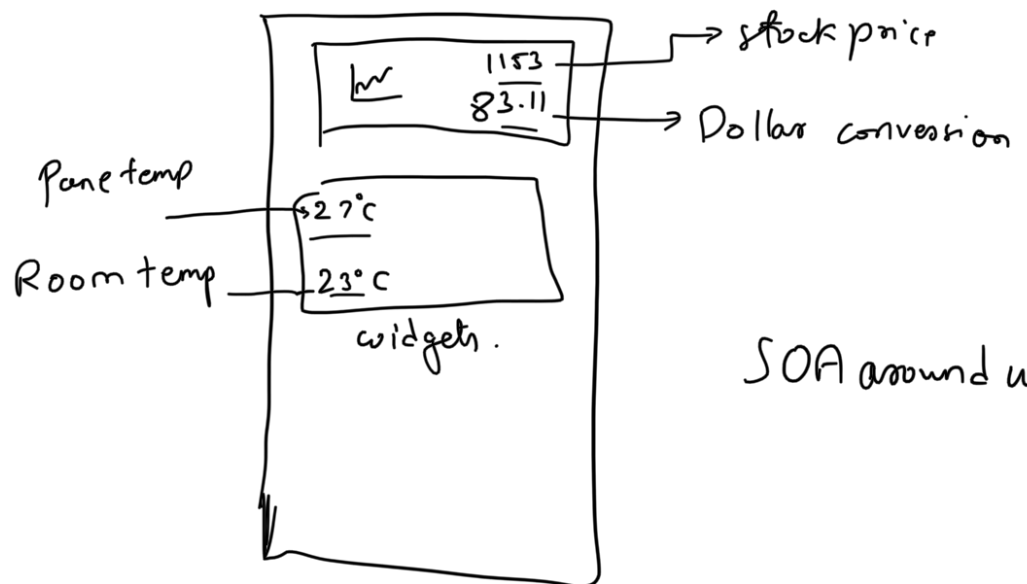
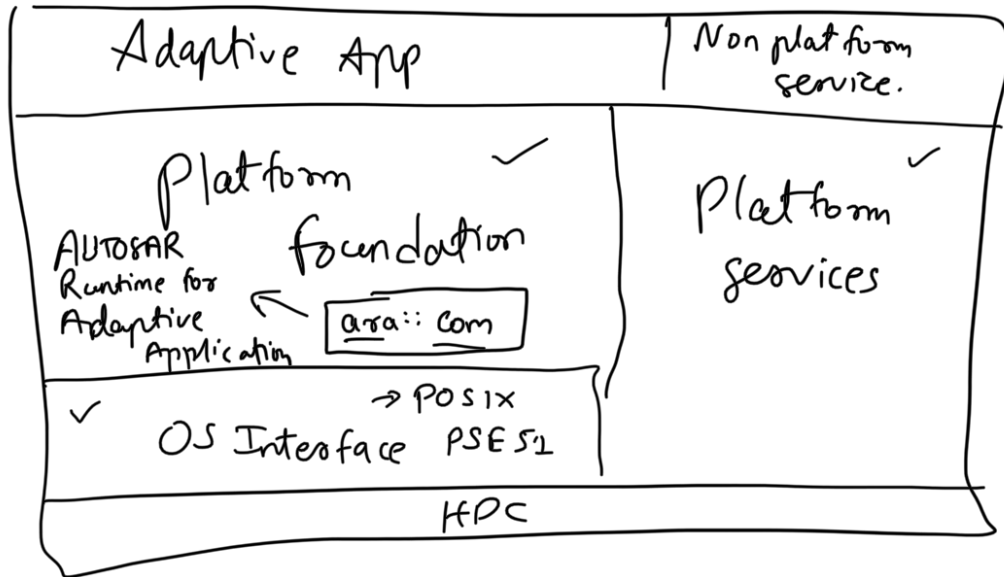
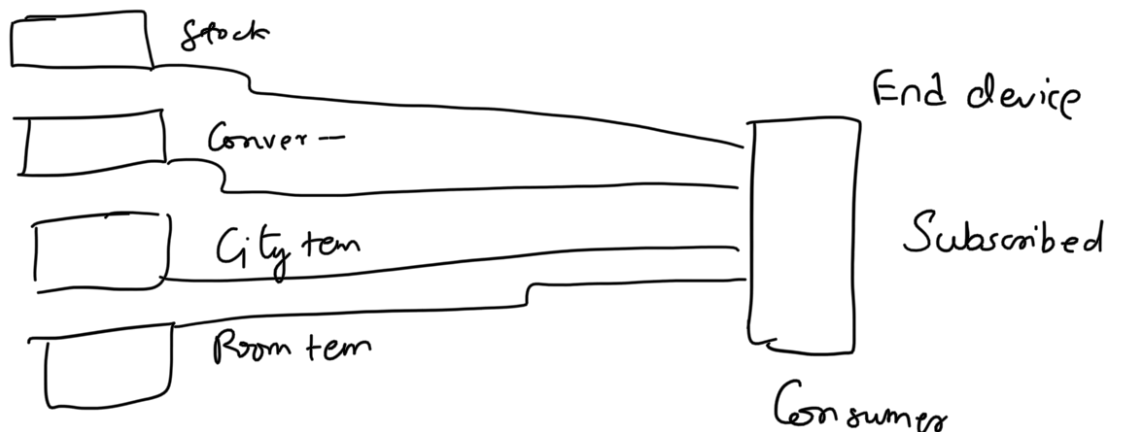


Service-Oriented Communication ✓

Communication management.



SOA around us.



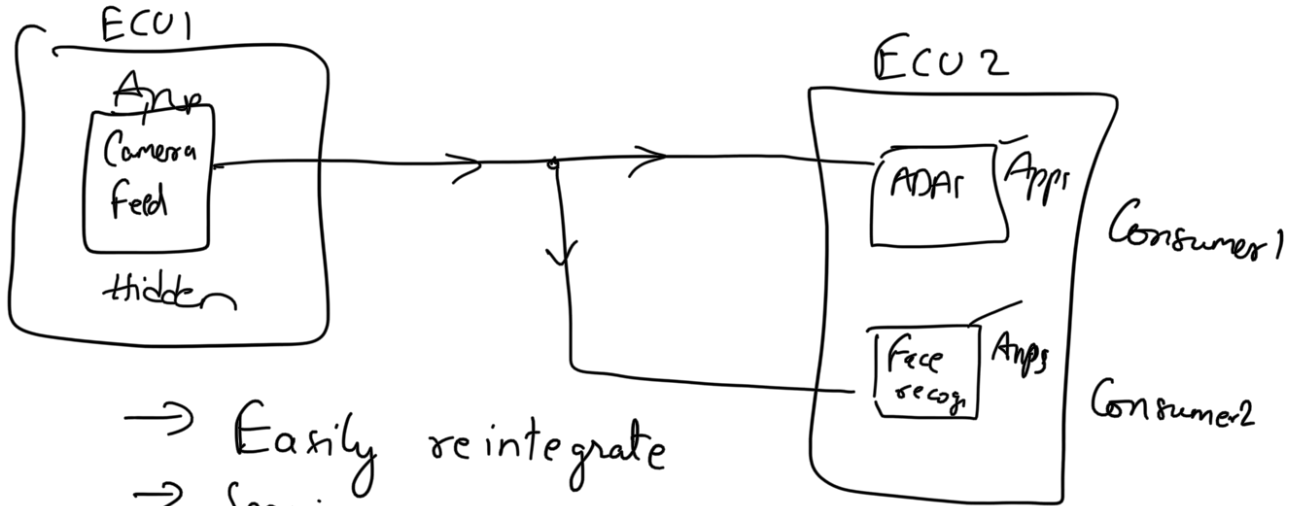
Feature

→ Automatic reconnection.

Service discoverability

→ Independent Services:

5 Adv



→ Easily reintegrate

→ Service reuse

→ Service Abstraction →

SOME/IP

Scalable service-Oriented Middleware over IP

OSI Model	Internet model RFC 1122	Middleware RFC 2768	
7 Application		Distributed Application	} SOME/IP
6 Presentation	Application	Middleware : Transport layer.	
5 Session			
4 Transport	Transport	TCP/UDP	
3 Network	Internet	IPv4 / IPv6	
2 Data link	Link	IEEE Ethernet MAC + VLAN	
1 Physical		Ethernet Phy.	✓

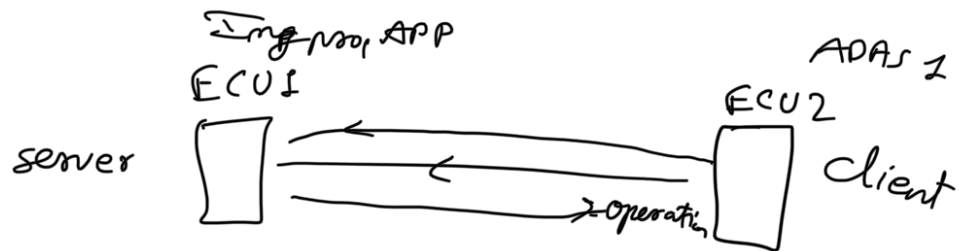
Service-Oriented communication ← SOME/IP / Comm mgmt / IPC
area: com

SOME/IP features → middleware features.

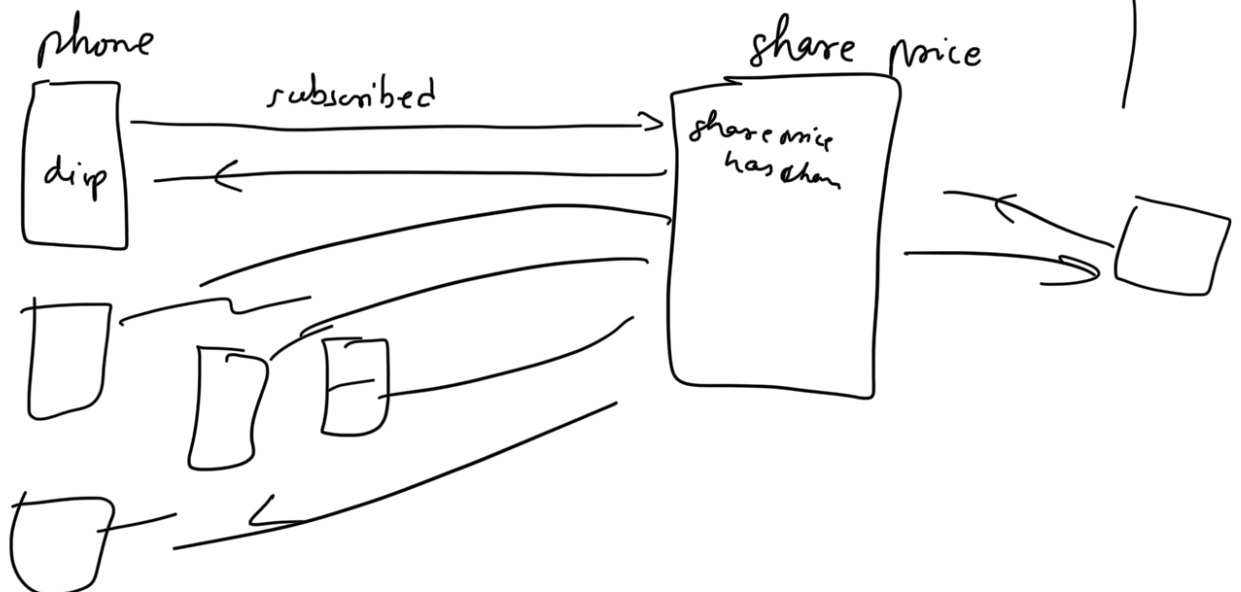
→ Serialization →



→ RPC → Remote Procedure call.

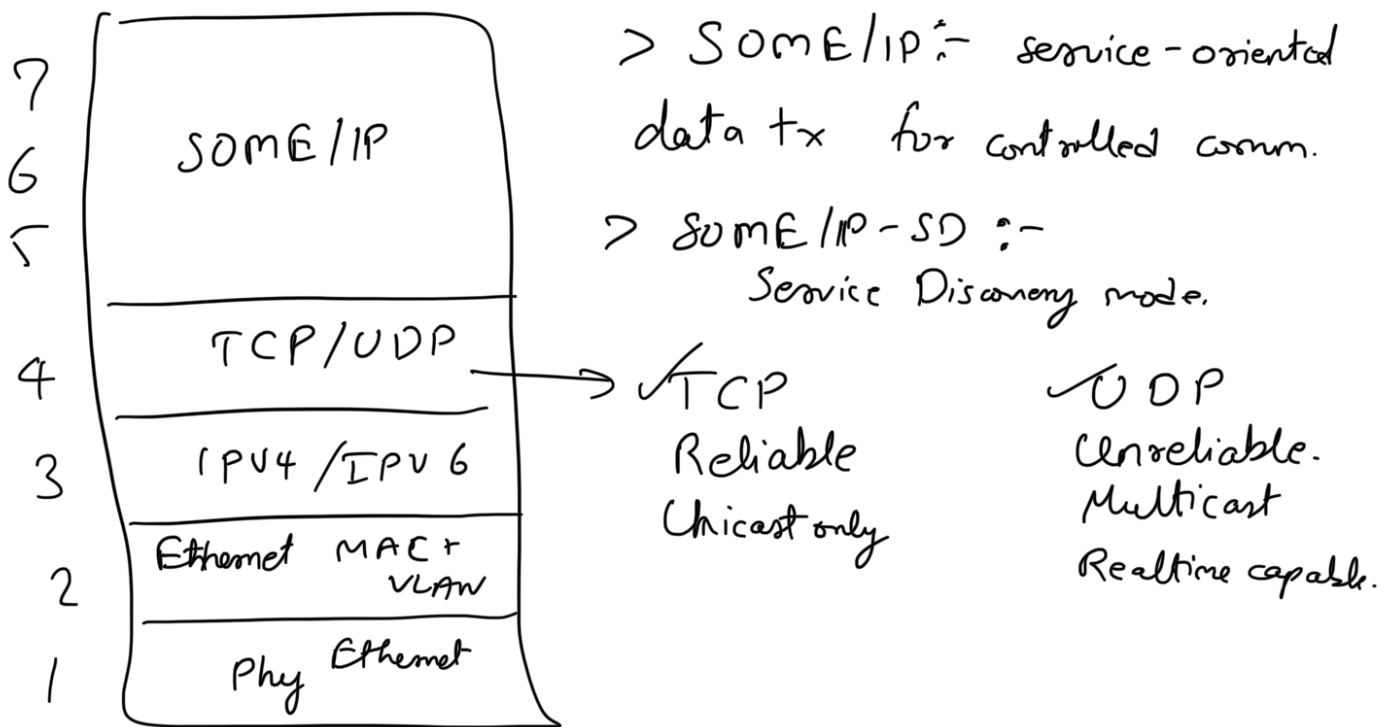


★ Pub/Sub (Publish / subscribe)



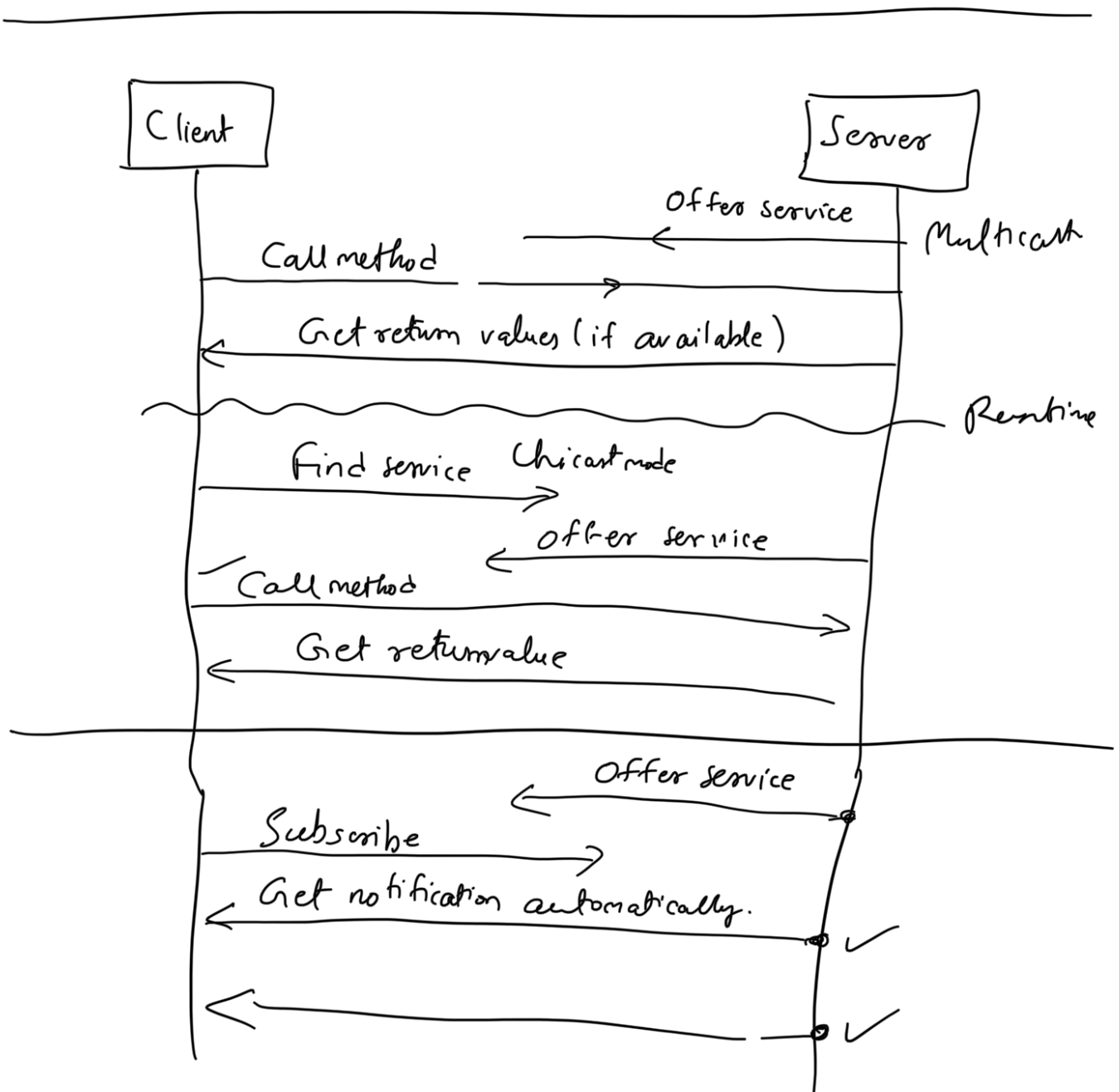
* Service Discovery - Dynamic finding of functionality

Overview



What's SOME/IP service features.

- * Methods - RPC
- * Events - Pub/Sub
- * fields -

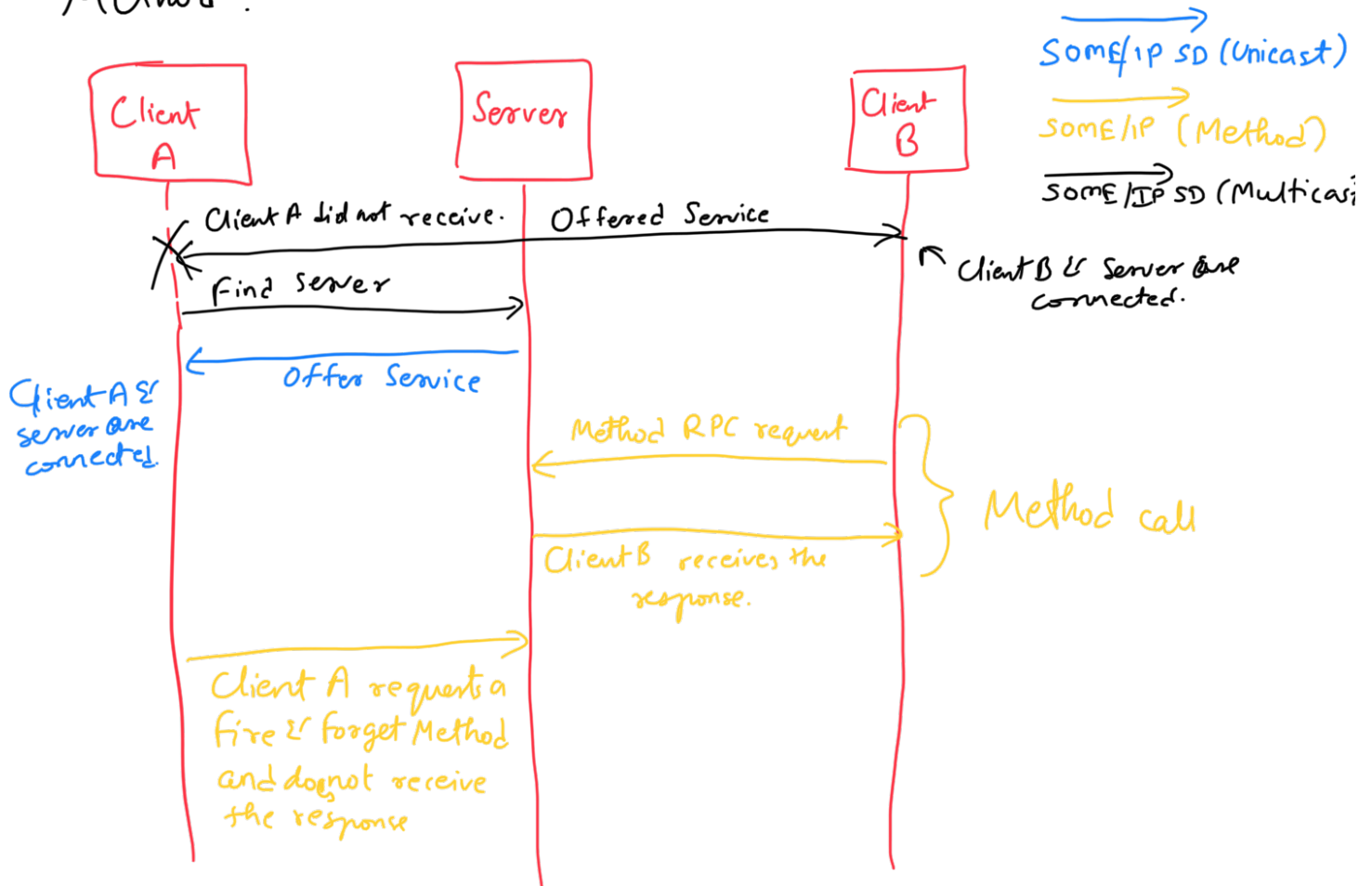


Adv

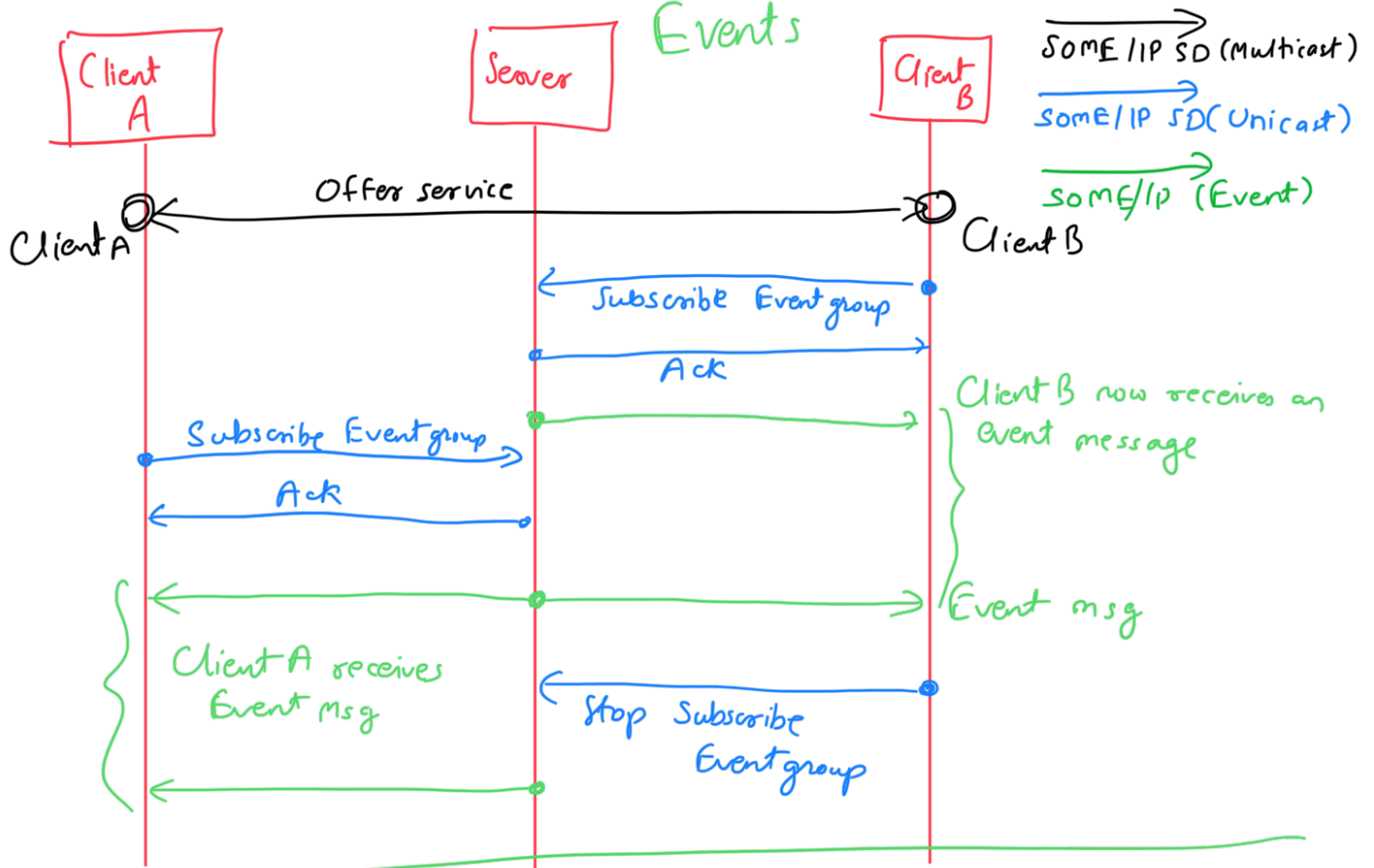
- Rx/Tx relations can be evaluated in ~~real~~ runtime.
- Easy relocation of service possible.
- Tx only with subscription.
minimize the CPU load.
- Save bandwidth

Protocol Flow → Methods/Events/Fields

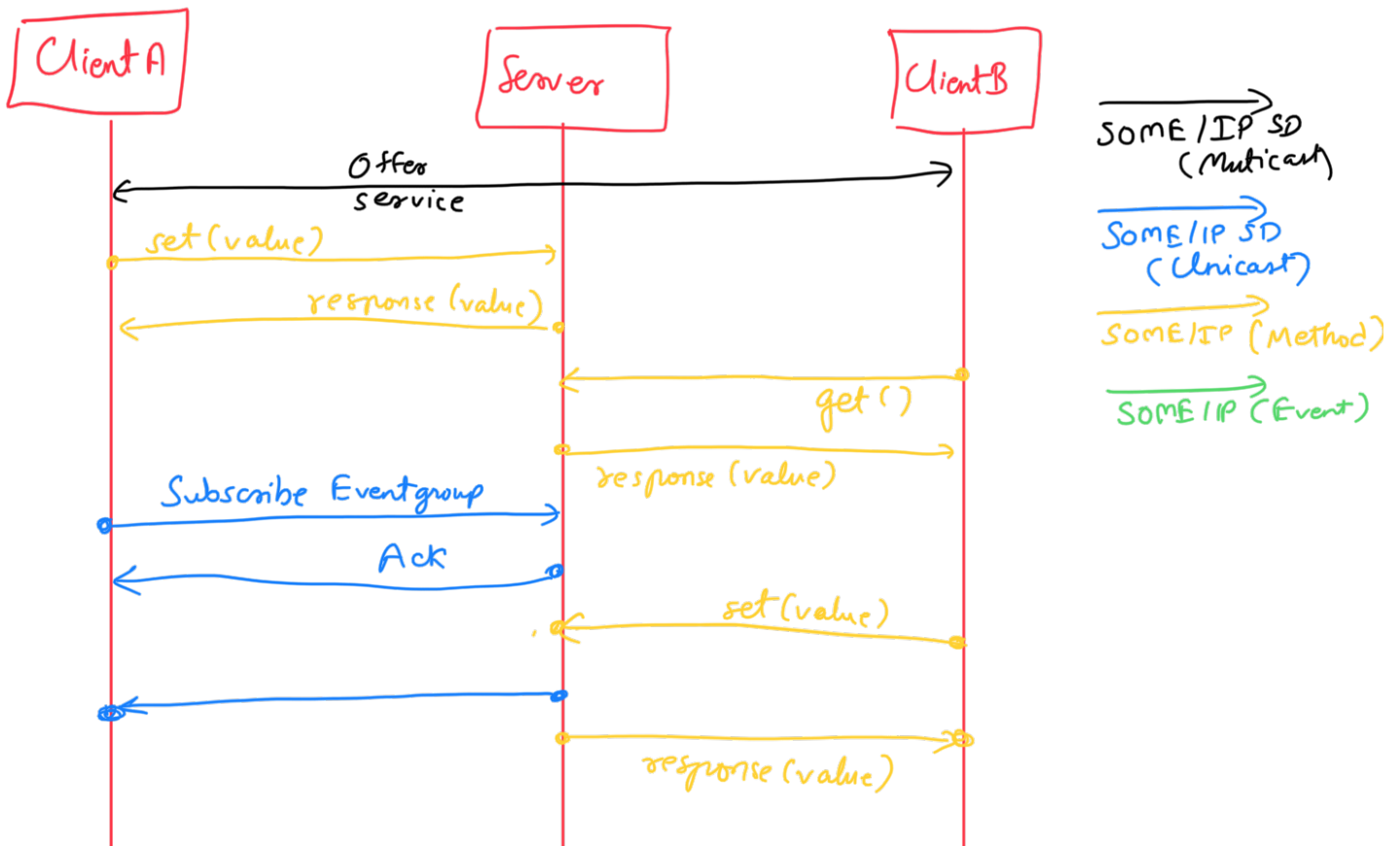
Method .



Events

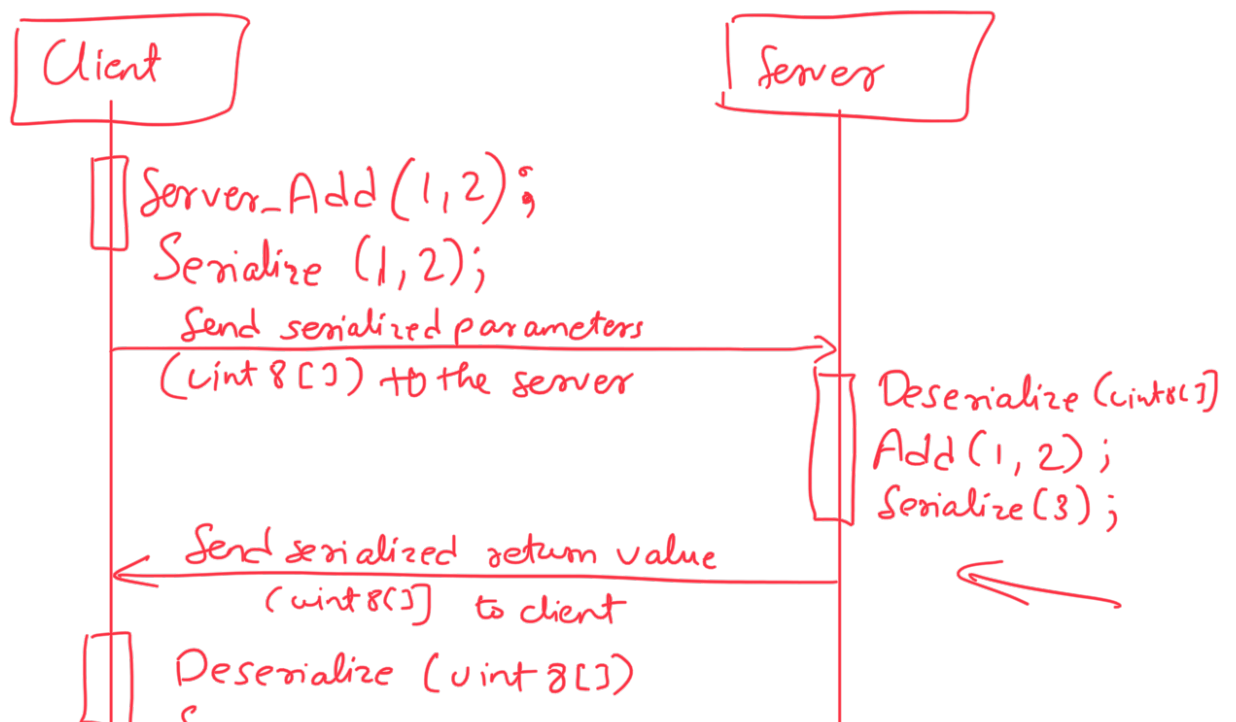
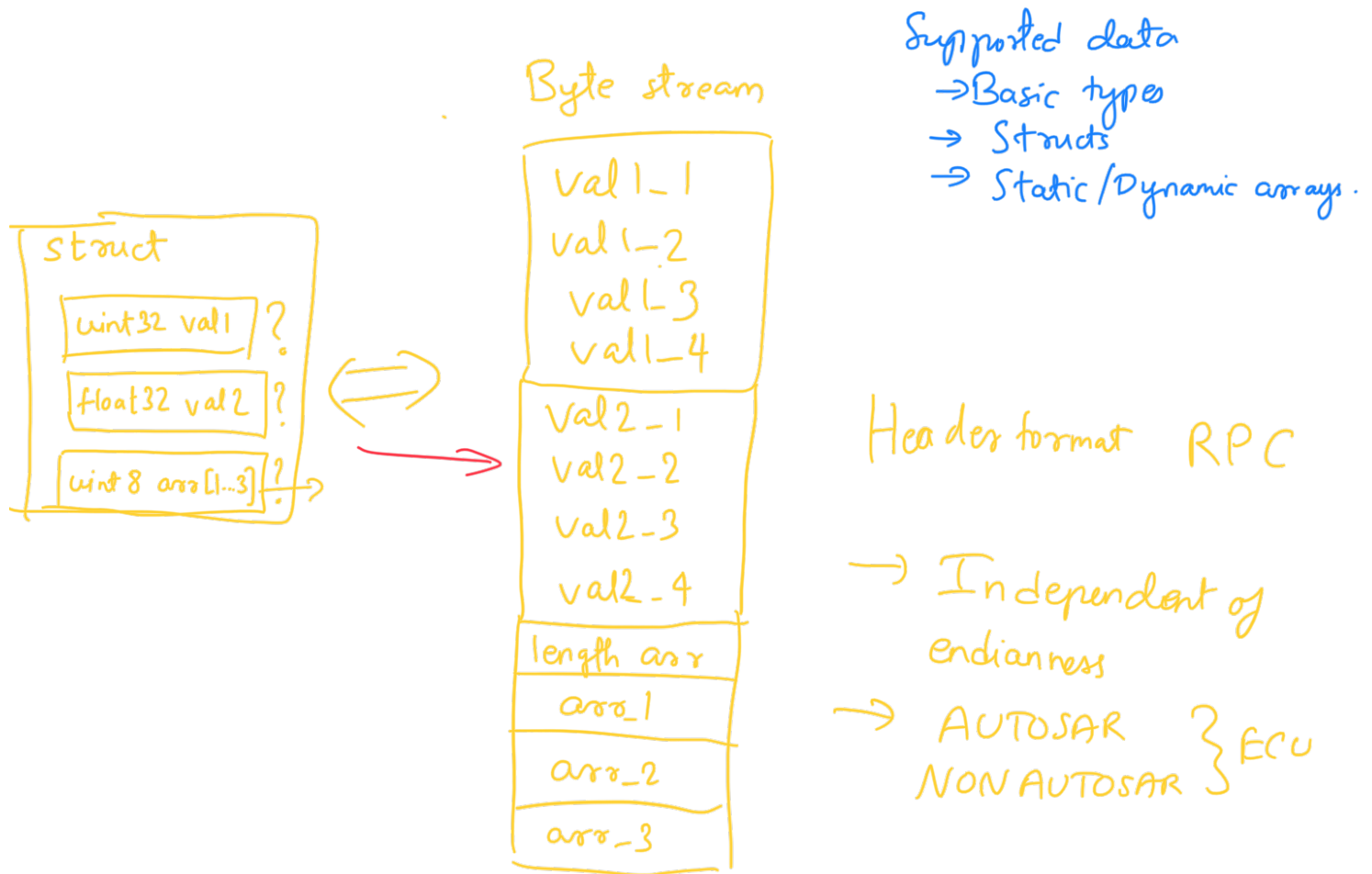


Fields



Data Serialization ✓

SOME/IP is an Automotive Serialization Protocol.



server_Acld_Result(3);

Data

→ Structure

→ UInt16

→ Variable lengthArray[3]

→ Structure

→ UInt8

→ Float32

→ UInt32 — 4 byte

→ Fixed Array[3]

→ SInt16 →

